pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Course Project

WEEK 4 · DAY 5

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The brief: the desk lead's note and the cover note

The Desk Lead's Note

Operations sent October's extract with a cover note. Before any number leaves the desk, tell me whether it can be trusted: fix what you can prove, in one function, so November is one call. Then tell me what is in the book, where it is concentrated, and how it differs from the benchmark. Send me a workbook and one page. Describe it: no views on any bond, issuer, or sector.

September's faults say nothing about October's, and every name is invented.

The Cover Note

```
To: Credit desk
From: Operations
Subject: Month-end holdings extract, as of 2026-10-30

The month-end extract is attached as project_holdings.csv.

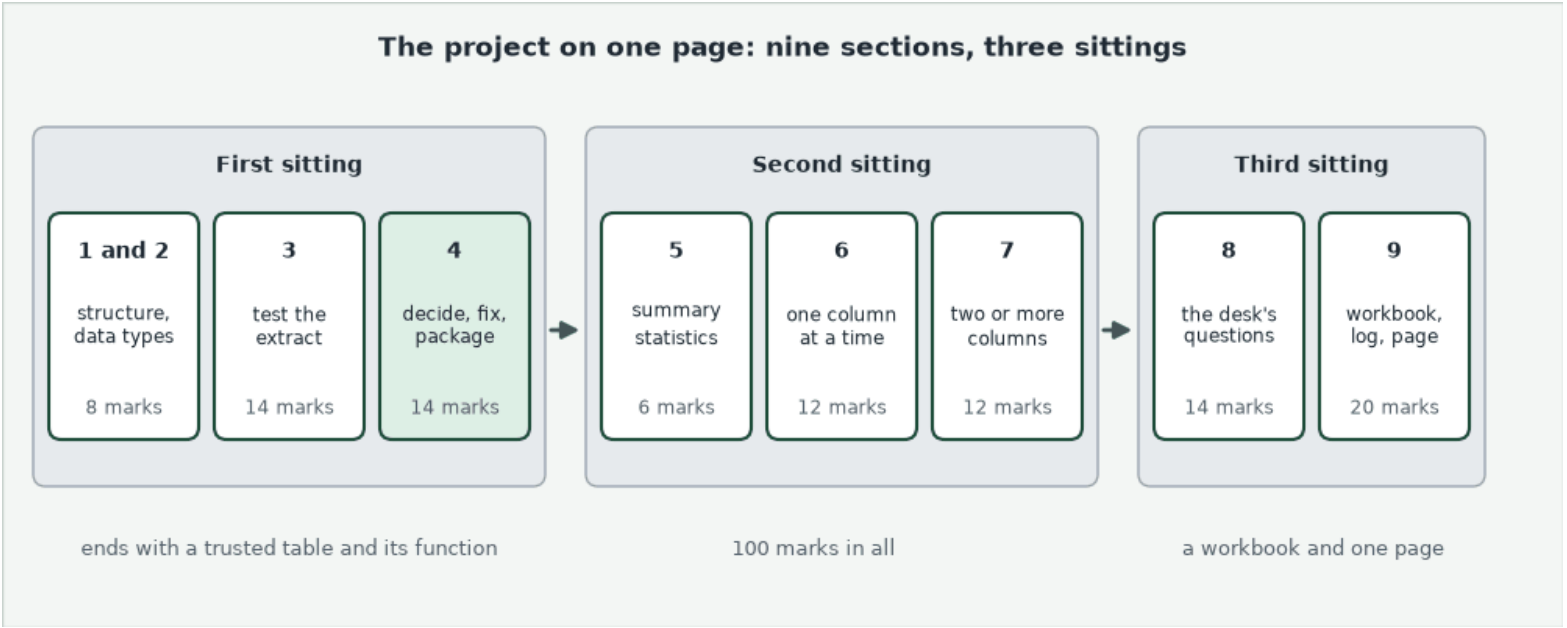
Control totals for the book:
  Number of bonds:      206
  Total face:           507,250,000 dollars
  Total market value:   509,928,477.50 dollars

Market value is face times clean price plus accrued interest, divided by 100.
All positions are priced as of 2026-10-30.
Please tell us if the extract does not agree with these totals.
```

It is also "data/project_cover_note.txt".

Before any number leaves the desk, tell me whether it can be trusted.

The project on one page: nine sections, three sittings



The extract ties to its cover note, and one call rebuilds it. The first sitting ends here. The second sitting, on "book" as at the checkpoint. The third sitting; a weight is a percent of the book's market value.

Nine sections, three sittings, and one page at the end. The file comes first.

The seven rules, and how the work is read

The Rules

1. Work in a copy: fixes go to "book", never to "raw".

2. State every decision in a markdown cell: what, how many rows, what evidence.

3. Print "headline()" before and after every fix.

4. Describe: never recommend, rank, or forecast.

5. Finish every chart: title, labelled axes, units.

6. Restart and Run All before you hand in.

7. Check any AI answer as week 1's Day 1, Working With an AI Assistant, taught: read, predict, run, check; log it for 9.2.

How the Work Is Read and Checked

The picture's marks show the weight; a number with no sentence earns part of its mark. Each Observations cell in sections 5 to 8 wants each figure with its unit and bond count. Each "check()" prints "correct", "not yet", or "not attempted yet"; round as each question asks. The checkpoint after 4.1 must pass before section 5. Worked answers: "course1_project_solutions.ipynb", once your page is written.

The data: units, one row, and the two other files

Units: "coupon" and "yield_pct" in percent, "oas" in basis points, "duration" in years, "price" and "accrued" per 100 of face, and "par_held", "amount_outstanding", and "market_value" in dollars.

One row of the extract: every column, its value, and its unit

as_of_date	2026-10-30	the date the book is valued
cusip	99080CAE8	the bond's identifier, nine characters
ticker	BEZE	the issuer's ticker
issuer	Bezane Finance	the issuer's name, invented
sector	Financials	a name from the benchmark's list
rating	BBB-	the bond's rating, AAA to CCC
coupon	7.0	annual coupon, percent
maturity	2032-06-18	the date the bond repays
price	106.576	clean price, per 100 of face
price_date	2026-10-30	the date of the price
accrued	2.567	accrued interest, per 100 of face
yield_pct	5.62	yield to maturity, percent
oas	149	option-adjusted spread, basis points
duration	4.545	modified duration, years
spread_duration	4.545	spread duration, years
dts	677.2	spread duration times OAS
amount_outstanding	1,650,000,000	face of the whole issue, dollars
par_held	4,000,000	face the desk holds, dollars
market_value	4,365,720.0	face times (price + accrued) / 100, dollars

4,000,000 x (106.576 + 2.567) / 100 = 4,365,720.00

"ratings.csv" is the rating scale: "rating", "score" (1 for AAA to 18 for CCC), "bucket", and "grade". "benchmark.csv" holds 821 bonds, a "weight_pct" for each, and no "par_held".

The benchmark is September's file, a month older than the book: say so wherever they meet.

1 The structure of the data; 2 data types

Tools: ".shape" and ".head()" (week 1's Day 5), "nunique()" (pandas Part 2).

1.1 Rows, Columns, and Identifiers

What shape is the extract (answer_1_1a), and what is one row? Show the first rows, and count the different as-of dates, CUSIPs (answer_1_1b), tickers, and issuers.

Write down the row count: the cover note says 206.

Tools: "dtypes" (pandas Part 1), "describe()" (week 1's Day 5), ".copy()" (pandas Part 2), "pd.to_datetime(errors='coerce')" (the EDA notebook), ".date()" (A First Look at the Data).

2.1 The Type of Each Column

Which types did pandas give the columns? How many are numbers (answer_2_1), and which hold dates?

2.2 A Working Copy With Real Dates

Copy "raw" to "book" and convert its three date columns. How many dates could not be read (answer_2_2)? Print each column's earliest and latest.

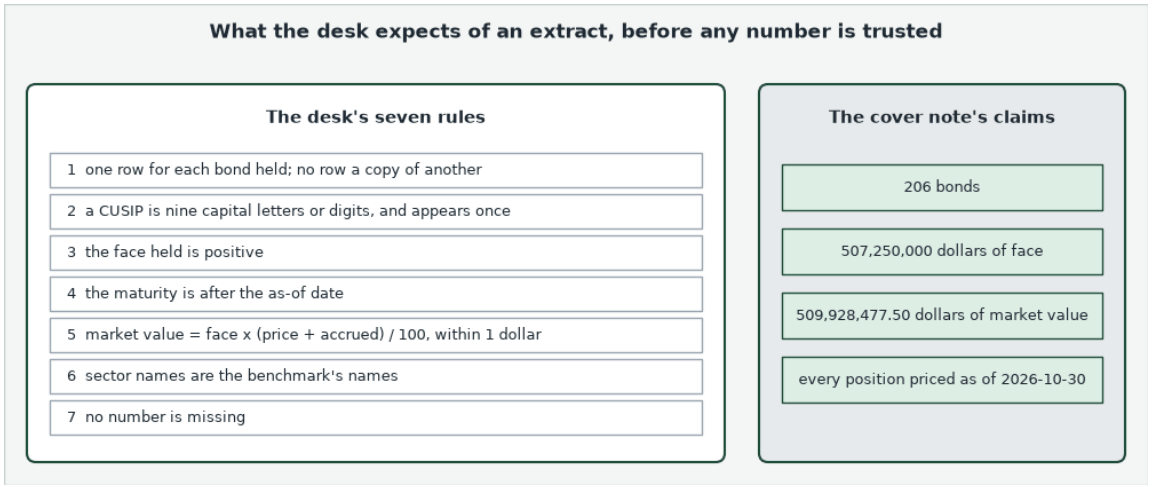
From here on every fix goes to "book".

3 Test the extract before fixing it

No list of faults comes with this file.

Test it against the cover note and the rules: fix what is an error, flag what you cannot fix, keep what is real.

Count on "book" as 2.2 left it, and fix nothing until section 4.



Excel side by side. CUSIP in B, sector E, price I, accrued K, OAS M, face R, value S.

Rule	Excel, any current version
1	=COUNTIF($B$2:$B$210, B2) > 1 marks rows sharing a CUSIP
2	=LEN(B2) = 9 tests only the length
3	=COUNTIF(R2:R210, "<=0")
5	=ABS(R2 * (I2 + K2) / 100 - S2) > 1 filled down
6	=COUNTIF(benchmark!$E$2:$E$822, E2) = 0 ignores capitals, so it misses names isin() catches
7	=COUNTBLANK(M2:M210)

The tests: count first, fix nothing until section 4

3.1 The File Against the Cover Note

Print the headline as received and the file less the cover note in rows (answer_3_1), face, and value. What do the three suggest?

3.2 Rule 1: One Row for Each Bond

How many rows copy an earlier row exactly (answer_3_2)? Show each beside its original, with the face and value they carry.

3.3 Rule 2: Identifiers

How many CUSIPs fail a pattern of nine capital letters or digits (answer_3_3a), and how many are not in the benchmark (answer_3_3b)? Is any CUSIP there twice with different values?

3.4 Rule 3: The Face Held

How many rows break rule 3 (answer_3_4)? Show them with price, accrued, face, and market value.

3.5 Rule 4: The Maturity

How many rows break rule 4 (answer_3_5)? Show them with coupon, maturity, and duration.

3.6 Rule 5: The Tie-Out

In how many rows is face times (price plus accrued) over 100 more than 1 dollar from "market_value" (answer_3_6)? Which column looks wrong in each?

3.7 Rule 6: Sector Names

How many rows have a "sector" that is not exactly a benchmark name (answer_3_7)? Print them so an end space would show.

3.8 Rule 7: Missing Numbers

How many cells are missing (answer_3_8a), and what share of the cover note's value are their rows, percent to 2 places (answer_3_8b)? What would dropping them do? Do the gaps share anything?

3.9 Price Dates and Six More Tests

How many rows are not priced on the as-of date (answer_3_9a)? Total the breaks of six tests (answer_3_9b): a price, coupon, or duration of zero or less; face above the amount outstanding; a rating off the scale; an untidy ticker.

4 Decide, fix, and package the cleaning

Tools: "drop_duplicates()", "fillna()" (the EDA notebook); ".loc[mask, column] =", "reset_index(drop=True)" (pandas Part 2); ".astype(int)" (the Treasury case study); ".map()" (the Consumer Credit case study); "def" (week 1's Day 1 Part 2); "assert" (week 1's Day 3); a function checked by its totals (From Notebook to Module).

4.1 Decide and Fix

Sort each failing row into the EDA notebook's three kinds: fix, flag, or keep. Give each fix a decision cell, then make it with "headline()" before and after. Rerun 3.2 to 3.9: how many rows still fail (answer_4_1)?

Checkpoint: the Cleaned Table

Does "book" match the cover note and the expected decisions? If a line is not "correct", open that part's Stuck block.

The Three Gaps, Closed

Did every gap to the cover note reach zero? This cell is written for you.

4.2 The Cleaning as a Function

Write "clean_extract(raw)", its docstring listing 2.2's dates and 4.1's fixes in order, leaving its input unchanged; a second table it needs is a second parameter. Store its result in cleaned, "assert" the three totals with messages, and show "cleaned" equals "book".

The extract ties to its cover note, and one call rebuilds it. The first sitting ends here.

5 Summary statistics; 6 one column at a time

The second sitting, on "book" as at the checkpoint.

5.1 The Rating Scale

Join the scale's "score", "bucket", and "grade" to "book", with rows before and after. How many bonds are High Yield (answer_5_1)?

5.2 Summary, Mean, Median, and Skew

Summarize "coupon", "price", "yield_pct", "oas", "duration", and "market_value": could a bond have each range? Give the median OAS to 1 place (answer_5_2a). Add "dts" and show mean, median, and skew: which column has the longest right tail, and which leans left? Give the OAS skew to 2 places (answer_5_2b).

5.3 Counts by Category

Count bonds by sector, by bucket from AAA to CCC, and by grade, drawn as three labelled bar charts in a row. Which bucket is largest (answer_5_3)?

Counts treat every bond alike; section 8 weighs them by value.

6.1 The Distribution of the OAS

Draw a histogram of the OAS with labelled lines at the mean and median. How many bonds sit above the mean (answer_6_1)?

6.2 The IQR Rule

Print the OAS quartiles, IQR, and limits, and draw its box plot. Give the upper limit to 3 places (answer_6_2a) and the bonds above it (answer_6_2b).

6.3 Errors or Real Bonds?

Show the flagged bonds widest first with rating, coupon, price, yield, and OAS. Errors or real? Give the evidence and your decision. Give their share of value, percent to 2 places (answer_6_3).

6.4 The Distribution of the Price

Draw a histogram of the clean price with a line at par. How many are below 100 (answer_6_4)? Is the lowest or highest priced bond an error?

6.5 The Size of the Positions

Draw a histogram of market value in millions of dollars. What share is the largest position, percent to 2 places (answer_6_5)? Does the IQR rule flag any?

7 Two or more columns

Tools: ".corr()", "sns.heatmap(annot=)", "sns.scatterplot(hue=)" (the Visualization notebook); ".loc[list]" (Combining Part 1); "sns.boxplot()" (the Trade Blotter case study); "pivot_table()" (Combining Part 2).

7.1 Correlations

Draw the correlations of "coupon", "price", "yield_pct", "oas", "duration", and "score" as an annotated heatmap. Name the strongest pair and one near zero. What does OAS against score mean (answer_7_1, 2 places)?

7.2 Yield Against Duration, by Grade

Draw yield against duration, coloured by grade. Give their correlation within Investment Grade to 2 places (answer_7_2): what does the whole-book figure hide?

7.3 OAS by Rating Bucket

Show each bucket's bond count and median, lowest, and highest OAS in credit order, and draw the medians as bars. Give the BBB median to 1 place (answer_7_3).

7.4 OAS by Grade

Draw one OAS box per grade, and print each grade's count and median. Give the High Yield median less the Investment Grade median to 1 place (answer_7_4).

7.5 Sector by Grade

Pivot market value in millions of dollars, sectors down and grades across, draw it as a heatmap, and check the cells add to the book. Which sector holds the most High Yield value (answer_7_5)?

A whole-book figure can hide what each grade does.

8 The desk's questions: weights, issuers, averages

The third sitting; a weight is a percent of the book's market value.

Tools: weights and the top ten (the Joining the Tables case study), weighted averages and "/ 365.25" (Combining Part 2), a signed bar (the Trade Blotter case study), a mean of 0s and 1s (the Consumer Credit case study).

8.1 Weight by Sector, Bucket, and Grade

Add "weight_pct" and check it adds to 100. Show bonds, value, and weight by sector, largest first, then by bucket and grade. Give the largest sector weight (answer_8_1a) and the High Yield weight (answer_8_1b) to 2 places.

8.2 The Ten Largest Issuers

How many issuers are there, and which ten hold the most market value? What share are the ten, percent to 2 places (answer_8_2)?

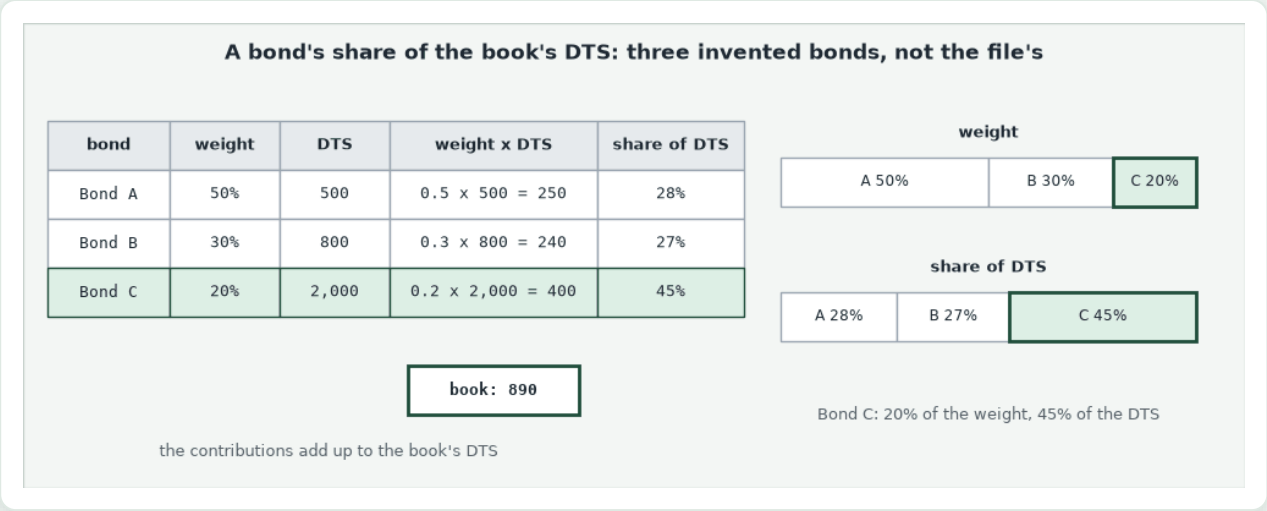
8.3 Weighted Averages

Give the average coupon, OAS, duration, and score, simple and weighted by market value. Give the weighted OAS to 1 place (answer_8_3a) and duration to 2 (answer_8_3b).

Excel side by side. The corrected book, rows 2 to 207, DTS in column P; any current version.

Figure	Excel
A sector's weight	=SUMIF(E2:E207, "Materials", S2:S207) / SUM(S2:S207) * 100, or a PivotTable value shown as % of Grand Total
Weighted OAS	=SUMPRODUCT(M2:M207, S2:S207) / SUM(S2:S207)
The book's DTS	=SUMPRODUCT(P2:P207, S2:S207) / SUM(S2:S207)

A bond's share of DTS, and what matures soon



A bond's DTS contribution is its weight as a fraction times its DTS; the contributions add up to the book's DTS.

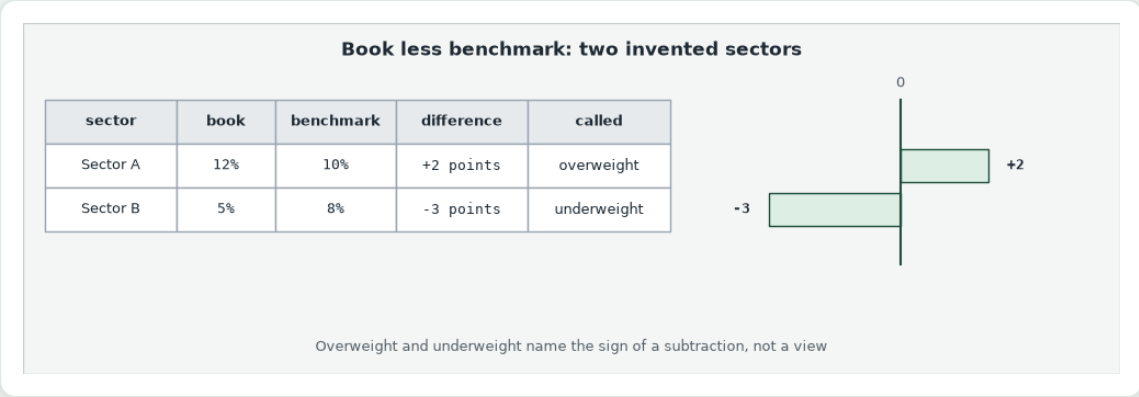
8.4 DTS by Sector

Add "dts_contribution", then show each sector's weight, contribution, and share of the book's DTS, largest first, and the same by grade. Give the book's DTS to 1 place (answer_8_4).

8.5 Maturing Within Three Years

Counting years as days to maturity over 365.25, what share of the book matures within three years, percent to 2 places (answer_8_5)?

Against the benchmark, DTS leaders, High Yield



A benchmark sector weight sums its bonds' "weight_pct". Positive is an overweight and negative an underweight: the sign of a subtraction, never a view.

8.6 Book Less Benchmark, by Sector

Show each sector's book weight, benchmark weight, and difference in percentage points, and draw the differences as bars with a zero line and a title naming both files' months. Give the largest positive difference's sector (answer_8_6a) and size to 2 places (answer_8_6b).

8.7 Book Less Benchmark, by Bucket

Join the scale to the benchmark, then compare by bucket in credit order and by grade, with the bucket differences drawn as bars. Give the BBB difference to 2 places (answer_8_7).

8.8 The Ten Largest DTS Contributions

Show the ten largest DTS contributions with CUSIP, ticker, rating, OAS, duration, DTS, and weight. Give their share of the book's DTS, percent to 1 place (answer_8_8), beside their weight.

8.9 High Yield Share of Bonds, by Sector

Add "is_hy", 1 for High Yield and 0 otherwise, and show each sector's bonds and High Yield share of bonds, highest first. Give the top share to 1 place (answer_8_9a) and its bond count (answer_8_9b).

Weights say where the dollars are; DTS shares say where the spread risk is.

9 The deliverables, and what to hand in

9.1 The Workbook

Write "project_report.xlsx": a "book" sheet without row labels and a "summary" sheet of 8.1's sector table. Read both back into answer_9_1a and answer_9_1b as row counts, "assert" the book sheet's value, delete the file, and store whether it exists in answer_9_1c.

9.2 The AI-Use Log

One row per question you asked an AI assistant; "none used" is valid.

9.3 The Page

Write the page for the portfolio manager below, 250 to 350 words, in five parts: the book in five numbers; concentrations by sector, grade, and issuer; data quality, each problem with its rows and its fix, and what was clean; the benchmark, with both files' dates; and open items for operations.

Nothing from an employer, no client names, positions, or credentials, goes into an AI tool or Colab: describe the columns, not the rows, and follow your firm's policy first.

Hand in this notebook, run from the top, with
every check "correct" and "raw" unchanged;
a decision cell and headlines for every fix;
"clean_extract" with its docstring and passing "assert"s;
no ".xlsx" file left behind, and every chart finished;
the log and the page.

The planted problems are in "data/project_answer_key.md": read it after you finish.

The note asked	Where
Can the file be trusted, and November in one call	sections 3 and 4, clean_extract
What is in the book, and how it differs from the benchmark	sections 5 to 8
A workbook and one page	9.1 and 9.3

Next is Course 2, Bond Math: prices, yields, duration, and DTS built by hand in your own "bondmath.py", as you kept "desk_tools.py".